

# Mohamed Khalil ZRELLY

Generative AI Engineer Apprentice | Computer Science Engineering Student

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## Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, specializing in agentic workflow orchestration. Computer Science Engineering student at Sup Galilée, Sorbonne Paris North University, with hands-on expertise in RAG, NLP, and reinforcement learning.

## Work Experience

### Crédit Agricole Personal Finance & Mobility

France

#### Generative AI Engineer Apprentice

September 2025 — Present · 10 months

- Developed a declarative, config-driven microservice for **workflow management and orchestration**, making it easy to manage and execute complex AI workflows at scale.
- Built an automated **workflow evaluation engine** leveraging **LLM-as-a-judge** to assess outputs against custom metrics, with a **dashboard** to analyse workflow behaviour.

Agentic Workflows · Advanced RAG · Amazon Bedrock · LLM Observability · Python/FastAPI · Amazon DynamoDB · Python/Streamlit

### VERMEG for Banking & Insurance Software

Tunisia

#### Natural Language Processing Intern

February 2024 — June 2024 · 5 months

- Implemented a system that automates email classification and information extraction from attachments, using fine-tuned **BERT** and **OCR** to improve accuracy and optimise efficiency.
- Developed a web application for **BERT Fine-Tuning**, email management, and real-time data analysis and visualisation, enabling model experimentation and performance monitoring.

Hugging Face Transformers · Sentence-Transformers · Fine-Tuning · PyTorch · Tesseract OCR · Python/Flask · MongoDB · React/Redux

### Sofiatech

Tunisia

#### Machine Learning Intern

June 2023 — August 2023 · 3 months

- Evaluated multiple **reinforcement learning** algorithms from **Stable Baselines 3** across various **Gymnasium** environments to assess agent performance and training efficiency.
- Built a **2D game** in **Pygame** where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas · Matplotlib · Python/Pygame

## Featured Projects

### CVision | RAG · CV Recommendation System

June 2025 — August 2025

Created an intelligent **CV recommendation system** based on job description using **Retrieval-Augmented Generation (RAG)**, to optimise the selection and automatic ranking of the most relevant profiles.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Milvus · Python/Streamlit

### Twixel | Multi-Agent · Social Listening System

In Development — 2026

Designing an intelligent **multi-agent system** for automated social listening on X (Twitter), aimed at identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · GPT-OSS · Qdrant · Python/FastAPI · Vue.js

## Technical Skills

|                                 |                                                              |
|---------------------------------|--------------------------------------------------------------|
| Generative AI & Agentic Systems | LangChain, LangGraph, Milvus, Qdrant, Amazon Bedrock         |
| LLM Evaluation & Observability  | DeepEval, Promptfoo, MLflow, Langfuse, Amazon CloudWatch     |
| Machine Learning, DL & MLOps    | PyTorch, TensorFlow, Scikit-learn, XGBoost, Amazon SageMaker |
| Data Analysis & Engineering     | Pandas, Matplotlib, PySpark, Apache Airflow, Amazon S3       |
| SQL & NoSQL Databases           | Oracle, MySQL, PostgreSQL, MongoDB, Amazon DynamoDB          |
| Full-Stack Web Development      | FastAPI, Flask, Express.js, Laravel, React, Vue.js           |
| DevOps & Cloud Infrastructure   | Docker, Amazon ECS, Fargate, Terraform, GitHub Actions       |
| Programming Languages (PL)      | Python, JavaScript/TS, PHP, Java, C/C++, PL/SQL              |

## Education

### Sup Galilée – Sorbonne Paris North University

Villetaneuse, France

#### Engineering Degree in Computer Science

2024 — 2027 (Expected)

### ISI Ariana – University of Tunis El Manar

Ariana, Tunisia

#### Bachelor of Science in Computer Science

2021 — 2024 · High Distinction

## Certifications & Languages

|                  |                                                                                         |
|------------------|-----------------------------------------------------------------------------------------|
| Microsoft Azure  | Azure AI Fundamentals (AI-900) — Jan 2024 · Azure Data Fundamentals (DP-900) — Jan 2024 |
| Udemy            | Spark and Python for Big Data with PySpark — Mar 2024 · NLP with Python — Feb 2024      |
| Spoken Languages | Arabic (Native) · French B2 — TCF · English B2 — TOEIC                                  |